

Database Tuning

Normalization

Functional Dependencies



Case ❶ ❷ ❸ ❹ | FD | Entailment | Equivalence

Employees

The Employee Table

The table **employee** record the salaries of the different employees in our organization. An agreement with the trade unions imposes that salaries are determined by the position. The actual value has been negotiated and fixed. The salary of a clerk is \$2000 per month, the salary of a manager is \$3000 per month, *etc.*

Employees

number	name	department	position	salary
1XU3	Dewi Srijaya	sales	clerk	2000
5CT4	Axel Bayer	marketing	trainee	1200
4XR2	John Smith	accounting	clerk	2000
7HG5	Eric Wei	sales	assistant manager	2200
4DE3	Winnie Lee	accounting	manager	3000
8HG5	Sylvia Tok	marketing	manager	3000

Case 1 2 3 4 | FD | Entailment | Equivalence

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8HG5	Sylvia Tok	marketing	manager	3000

Dependencies

Salaries are **determined** by position.

Functional

This kind of business rule can be translated into an integrity constraint called **functional dependencies**.

{position} → {salaries}

Case 1 2 3 4 | FD | Entailment | Equivalence

Employees

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Employees

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7HG5	Eric Wei	sales	assistant manager	2200
4DE3	Winnie Lee	accounting	<u>manager</u>	3000
8HG5	Sylvia Tok	marketing	<u>manager</u>	4000

Dependencies

Salaries are **determined** by position.

Disallowed

Given **{position} → {salaries}**,
the following instance is an **invalid** instance.

Case 1 2 3 4 | FD | Entailment | Equivalence

Employees

The Employee Table

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Employees

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5CT4	Axel Bayer	marketing	trainee	1200
4XR2	John Smith	accounting	clerk	2000
7HG5	Eric Wei	sales	assistant manager	2200
4DE3	Winnie Lee	accounting	<u>manager</u>	3000
8HG5	Sylvia Tok	marketing	<u>senior manager</u>	3000

Dependencies

Salaries are **determined** by position.

Allowed

Given **{position} → {salaries}**,
the following instance is a **valid** instance.

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Definition

Functional Dependencies

An **instance** r (a table) of a relation schema R satisfies the **functional dependencies** σ or the form $X \rightarrow Y$ with $X \subseteq R$ and $Y \subseteq R$

if and only if

if two tuples of r **agree** on their X -values, then they agree on their Y -values.

$$r \models X \rightarrow Y \quad \leftrightarrow \quad \forall t1 \in r : \forall t2 \in r : ((t1.X = t2.X) \Rightarrow (t1.Y = t2.Y))$$

Terminology

$X \rightarrow Y$ reads

X **functionally determines** Y

X **determines** Y

X **implies** Y

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

SQL

Checking

The following query checks if $\{\text{position}\} \rightarrow \{\text{salary}\}$ functional dependency constraint is violated.

```
SELECT * FROM employees e1, employees e2
WHERE e1.position = e2.position AND e1.salary <> e2.salary;
```

Enforcement??

The following SQL **CHECK** constraints enforces the functional dependency.

```
CHECK NOT EXISTS (
  SELECT * FROM employees e1, employees e2
  WHERE e1.position = e2.position AND e1.salary <> e2.salary)
```

However, we cannot use SQL query in **CHECK** constraints.

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

SQL

Checking

The following query checks if $\{\text{position}\} \rightarrow \{\text{salary}\}$ functional dependency constraint is violated.

```
SELECT * FROM employees e1, employees e2
WHERE e1.position = e2.position AND e1.salary <> e2.salary;
```

Enforcement!

The following SQL trigger function and trigger enforce the functional dependencies

```
CREATE OR REPLACE FUNCTION fd()
RETURNS TRIGGER AS $$ BEGIN
  IF EXISTS ( SELECT *
    FROM employees e1, employees e2
    WHERE e1.position = e2.position
      AND e1.salary <> e2.salary )
    THEN RAISE EXCEPTION 'fd!'; END IF;
  RETURN NEW;
END; $$ LANGUAGE plpgsql;
```

```
CREATE CONSTRAINT TRIGGER tr
AFTER INSERT OR UPDATE OR DELETE ON employees
DEFERRABLE INITIALLY DEFERRED
FOR EACH ROW
EXECUTE FUNCTION fd();
```

However, this is not efficient.

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

SQL

Checking

The following query checks if $\{\text{position}\} \rightarrow \{\text{salary}\}$ functional dependency constraint is violated.

```
SELECT * FROM employees e1, employees e2
WHERE e1.position = e2.position AND e1.salary <> e2.salary;
```

Enforcement!?

We **may** enforce functional dependency when it corresponds to a **key** of the table by declaring **PRIMARY KEY** or **UNIQUE** constraint.

```
CREATE TABLE employees (
  -- code omitted
  position VARCHAR(32) PRIMARY KEY,
  -- code omitted
);
```

However, it does not work if **position** is not a key. This may be achieved via decomposition.

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Quiz

Question

Consider $R = \{A, B, C, D\}$. Which of the following 4 relation instances (*i.e., tables*) is **valid** for the functional dependency $\{A, B\} \rightarrow \{D\}$?

Select all options that apply.

Option A

A	B	C	D

Option B

A	B	C	D
1	2	a	4
1	3	b	4
1	2	c	3

Option C

A	B	C	D
1	3	a	4
1	2	b	3
1	3	c	4

Option D

A	B	C	D
1	2	a	4
1	2	b	3

Choice	Comment	
A Option A	YES: empty is always valid	✓
B Option B	YES: no violation	✓
C Option C	NO: row 1 and 3 is a violation	✗
D Option D	NO: row 1 and 2 is a violation	✗

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Quiz

Question

Consider an instance of the **employees** table. Which of the following functional dependency is **definitely** violated?

number	name	department	position	salary
1XU3	Dewi Srijaya	sales	clerk	2000
5CT4	Axel Bayer	marketing	trainee	1200
4XR2	John Smith	accounting	clerk	2000
7HG5	Eric Wei	sales	assistant manager	2200
4DE3	Winnie Lee	accounting	manager	3000
8HG5	Sylvia Tok	marketing	manager	3000

Choice	Comment
A {name,salary} → {department}	NO: no known violation 
B {name} → {number}	NO: no known violation 
C {department} → {position}	YES: row 1 and 4 is a violation 
D {position, salary} → {salary}	NO: cannot be violation 

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Definition

Trivial, Non-Trivial, Completely Non-TrivialA functional dependency $X \rightarrow Y$ is**trivial**

if and only if

$$Y \subseteq X$$

non-trivial

if and only if

$$Y \not\subseteq X$$

completely non-trivial

if and only if

$$Y \neq \emptyset \quad \underline{\text{and}} \quad Y \cap X = \emptyset$$

Note

Our convention allows for any of the following

$$\emptyset \rightarrow \emptyset$$

$$\emptyset \rightarrow Y$$

$$X \rightarrow \emptyset$$

(Non-)Trivial

The definition of non-trivial is quite literally the negation of trivial.

The definition of completely non-trivial requires the functional dependency to first be non-trivial.

Quiz

Trivial, Non-Trivial, Completely Non-Trivial

A functional dependency $X \rightarrow Y$ is

trivial	if and only if	$Y \subseteq X$
non-trivial	if and only if	$Y \not\subseteq X$
completely non-trivial	if and only if	$Y \neq \emptyset$ <u>and</u> $Y \cap X = \emptyset$

Question

Which of the following functional dependency is **non-trivial** but not **completely non-trivial**.

Choice	Comment	
A $\{A,C\} \rightarrow \{A,B,C\}$	YES: $\{A,B,C\} \not\subseteq \{A,C\}$ but has intersection	✓
B $\{A,C\} \rightarrow \{B\}$	NO: this is completely non trivial	✗
C $\{A,B\} \rightarrow \{A,B\}$	NO: this is trivial	✗
D $\{A\} \rightarrow \emptyset$	NO: this is trivial	✗

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Definition

Superkey, Candidate Key, Prime Attribute

Given a relation schema R and a set of functional dependencies Σ , a set of attribute $S \subseteq R$ is a

superkey	if and only if	$S \rightarrow R$
candidate key (<i>or just key</i>)	if and only if	$S \rightarrow R$ <u>and</u> for all $T \subset S : T \not\rightarrow R$

A **prime attribute** is an attribute that appears in at least one candidate key of R with Σ (*otherwise known as non-prime attribute*).

Minimal

The condition "for all $T \subset S : T \not\rightarrow R$ " means that $S \rightarrow R$ is a **minimal** superkey. In other words, S is a **candidate key** if and only if

S determines the value of the entire tuple, and
no proper subset of S can determine the value of the entire tuple

Case | **FD** 1 2 3 4 5 6 7 | Entailment | Equivalence

Quiz

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A **prime attribute** is an attribute that appears in at least one candidate key of R with Σ (otherwise known as non-prime attribute).

Question

Consider

$$R = \{A, B, C, D\}$$
$$\Sigma = \{ \{A, B\} \rightarrow \{C, D\}, \{C\} \rightarrow \{A, B\} \}$$

- What are the candidate keys?
- What are the prime attributes?
- How do we **compute** them?

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Definition

Logical Entailment

Let Σ be a set of functional dependencies of a relation schema R . Let $X \rightarrow Y$ be a functional dependency.

$X \rightarrow Y$ is **logically entailed** by Σ

if and only if

for all relation instance r satisfying all functional dependency $\sigma \in \Sigma$, r also satisfies $X \rightarrow Y$.

$$\Sigma \models X \rightarrow Y \quad \leftrightarrow \quad \forall r: (\forall \sigma \in \Sigma: r \models \sigma) \Rightarrow r \models X \rightarrow Y$$

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

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Case

$R = \{A, B, C, D\}$
 $\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$

Questions

Is $\Sigma \models \{C\} \rightarrow \{B\}$?

Is $\Sigma \models \{A,D\} \rightarrow \{B,C\}$?

Is $\Sigma \models \{C,D\} \rightarrow \{D\}$?

Is $\Sigma \models \{C,D\} \rightarrow \{B,A\}$?

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

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Case

$R = \{A, B, C, D\}$
 $\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$

Questions

Is $\Sigma \models \{C\} \rightarrow \{B\}$?

Is $\Sigma \models \{A,D\} \rightarrow \{B,C\}$?

Is $\Sigma \models \{C,D\} \rightarrow \{D\}$?

Is $\Sigma \models \{C,D\} \rightarrow \{B,A\}$?

Issues

Armed with only the definition of functional dependencies, the problems of checking entailment is a daunting task.

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Amstrong's Axioms

Inference Rules

Let R be a set of attributes. The following **inference rules** are the **Armstrong Axiom**.

- **Reflexivity**

$$\forall X \subseteq R : \forall Y \subseteq R : ((Y \subseteq X) \Rightarrow (X \rightarrow Y))$$

Trivial functional dependencies are always valid

- **Augmentation**

$$\forall X \subseteq R : \forall Y \subseteq R : \forall Z \subseteq R : ((X \rightarrow Y) \Rightarrow ((X \cup Z) \rightarrow (Y \cup Z)))$$

*You can add the same attribute to both left hand and right hand side **at the same time***

- **Transitivity**

$$\forall X \subseteq R : \forall Y \subseteq R : \forall Z \subseteq R : ((X \rightarrow Y \wedge Y \rightarrow Z) \Rightarrow (X \rightarrow Z))$$

You can short-circuit the arrows

Production Rule

We can represent functional dependencies as **production rules**. Armstrong's axioms derived other items that can be produced.

Notation

If a functional dependency $X \rightarrow Y$ can be derived using Armstrong's axioms from Σ , we denote this as

$$\Sigma \vdash X \rightarrow Y$$

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Examples

Case

- $R = \{A, B, C, D\}$
- $\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$

```
1 {A}
  ->
  {B}
  [G
  ive
  n]
```



- [Reflexivity]

$$\forall X \subseteq R: \forall Y \subseteq R: ((Y \subseteq X) \Rightarrow (X \rightarrow Y))$$
- [Augmentation of (<line>) with {<attr>}]

$$\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R: \\ ((X \rightarrow Y) \Rightarrow ((X \cup Z) \rightarrow (Y \cup Z)))$$
- [Transitivity of (<line>) and (<line>)]

$$\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R: \\ ((X \rightarrow Y \wedge Y \rightarrow Z) \Rightarrow (X \rightarrow Z))$$

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Soundness

Theorem

The **reflexivity** inference rule is sound (i.e., correct/valid).

The **augmentation** inference rule is sound (i.e., correct/valid).

The **reflexivity** inference rule is sound (i.e., correct/valid).

Soundness

$$\Sigma \vdash X \rightarrow Y \quad \Rightarrow \quad \Sigma \models X \rightarrow Y$$

If you can derive it, it is valid!

Theorems

Theorem

The **reflexivity** inference rule is sound (i.e., correct/valid).

The **augmentation** inference rule is sound (i.e., correct/valid).

The **reflexivity** inference rule is sound (i.e., correct/valid).

Proof of Reflexivity

- Let Σ be a set of functional dependencies on a relation schema R . Let $X \rightarrow Y$ and $Y \rightarrow Z$ be in Σ .
- We know for all valid instance r of R with Σ
 $(\forall t1 \in r : \forall t2 \in r : ((t1.X = t2.X) \Rightarrow (t1.Y = t2.Y)))$ (definition of $X \rightarrow Y$)
- We know for all valid instance r of R with Σ
 $(\forall t1 \in r : \forall t2 \in r : ((t1.Y = t2.Y) \Rightarrow (t1.Z = t2.Z)))$ (definition of $Y \rightarrow Z$)
- Therefore, for all valid instance r of R with Σ
 $(\forall t1 \in r : \forall t2 \in r : ((t1.X = t2.X) \Rightarrow (t1.Z = t2.Z)))$
- Therefore, $X \rightarrow Z$ by definition of FD.

Extension

Theorem

The **union** inference rule is sound (i.e., correct/valid).

$$\forall X \in R: \forall Y \in R: \forall Z \in R: (((X \rightarrow Y) \wedge (X \rightarrow Z)) \Rightarrow (X \rightarrow (Y \cup Z)))$$

1

- [Reflexivity]

$$\forall X \subseteq R: \forall Y \subseteq R: ((Y \subseteq X) \Rightarrow (X \rightarrow Y))$$

- [Augmentation of (<line>) with {<attr>}]

$$\begin{aligned} &\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R: \\ &((X \rightarrow Y) \Rightarrow ((X \cup Z) \rightarrow (Y \cup Z))) \end{aligned}$$

- [Transitivity of (<line>) and (<line>)]

$$\begin{aligned} &\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R: \\ &((X \rightarrow Y \wedge Y \rightarrow Z) \Rightarrow (X \rightarrow Z)) \end{aligned}$$



Extension

Theorem

The **decomposition** inference rule is sound (i.e., correct/valid).

$$\forall X \in R: \forall Y \in R: \forall Z \in R: ((X \rightarrow (Y \cup Z)) \Rightarrow ((X \rightarrow Y) \wedge (X \rightarrow Z)))$$

1 {X}
->
{Y
,Z}

[Gi
ven
]



- [Reflexivity]

$$\forall X \subseteq R: \forall Y \subseteq R: ((Y \subseteq X) \Rightarrow (X \rightarrow Y))$$
- [Augmentation of (<line>) with {<attr>}]

$$\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R:$$

$$((X \rightarrow Y) \Rightarrow ((X \cup Z) \rightarrow (Y \cup Z)))$$
- [Transitivity of (<line>) and (<line>)]

$$\forall X \subseteq R: \forall Y \subseteq R: \forall Z \subseteq R:$$

$$((X \rightarrow Y \wedge Y \rightarrow Z) \Rightarrow (X \rightarrow Z))$$

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Completeness

Theorem (Hard!)

The Armstrong's axioms are **complete**.

$$(\Sigma \models X \rightarrow Y) \Rightarrow (\Sigma \vdash X \rightarrow Y)$$

If it is valid, you can derive it!

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Completeness

Theorem (Hard!)

The Armstrong's axioms are **complete**.

$$(\Sigma \models X \rightarrow Y) \Rightarrow (\Sigma \vdash X \rightarrow Y)$$

If it is valid, you can derive it!

Theorem (Contrapositive)

The Armstrong's axioms are **complete**.

$$(\Sigma \not\models X \rightarrow Y) \Rightarrow (\Sigma \not\vdash X \rightarrow Y)$$

If you can not derive it, it is not valid!

Case | FD | **Entailment** 1 2 3 4 5 6 | Equivalence

Completeness

Theorem (Hard!)

The Armstrong's axioms are **complete**.

$$(\Sigma \models X \rightarrow Y) \Rightarrow (\Sigma \vdash X \rightarrow Y)$$

If it is valid, you can derive it!

Theorem (Contrapositive)

The Armstrong's axioms are **complete**.

$$(\Sigma \not\models X \rightarrow Y) \Rightarrow (\Sigma \not\vdash X \rightarrow Y)$$

If you can not derive it, it is not valid!

Implication

Soundness If you can derive it, it is valid.

Completeness If you can not derive it, it is not valid.

There is a **decision procedure** to check for a validity of a functional dependencies given Σ

Case | FD | Entailment | **Equivalence** ❶ ❷ ❸ ❹

Definition

Closure of Set of Functional Dependencies

Let Σ be a set of functional dependencies of a relation schema R . The **closure** of Σ is the set of functional dependencies logically entailed by the functional dependencies in Σ .

$$\Sigma^+ = \{X \rightarrow Y \mid \Sigma \models X \rightarrow Y\}$$

Find

- trivial functional dependencies in Σ^+ .
- non-trivial but not completely non-trivial functional dependencies in Σ^+ .
- completely non-trivial functional dependencies in Σ^+ .

Case | FD | Entailment | **Equivalence** ❶ ❷ ❸ ❹

Definition

Closure of Set of Functional Dependencies

Let Σ be a set of functional dependencies of a relation schema R . The **closure** of Σ is the set of functional dependencies logically entailed by the functional dependencies in Σ .

$$\Sigma^+ = \{X \rightarrow Y \mid \Sigma \models X \rightarrow Y\}$$

Attribute Closure

Let Σ be a set of functional dependencies of a relation schema R . The **closure** of the set of attribute $S \subseteq R$ is the set of all attributes functionally determined by S .

$$S^+ = \{A \in R \mid \exists (S \rightarrow \{A\}) \in \Sigma^+\}$$

Case | FD | Entailment | **Equivalence** ❶ ❷ ❸ ❹

Definition

Closure of Set of Functional Dependencies

Let Σ be a set of functional dependencies of a relation schema R . The **closure** of Σ is the set of functional dependencies logically entailed by the functional dependencies in Σ .

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Attribute Closure

Let Σ be a set of functional dependencies of a relation schema R . The **closure** of the set of attribute $S \subseteq R$ is the set of all attributes functionally determined by S .

$$S^+ = \{A \in R \mid \exists (S \rightarrow \{A\}) \in \Sigma^+\}$$

Implication

$S \rightarrow S^+$ *by definition of functionally determined*

$\forall A \in S^+ : S \rightarrow \{A\}$ *using decomposition rule*

- If $A \in S^+$, then $S \rightarrow \{A\}$
- If $A \notin S^+$, then $S \not\rightarrow \{A\}$

Case | FD | Entailment | **Equivalence** 1 2 3 4

Algorithm

Algorithm 1: Attribute Closure Algorithm**input:** S, Σ **output:** S^+ **begin** $\Omega := \Sigma$; *// unused* $\Gamma := S$; *// closure***while** $(X \rightarrow Y)$ and $(X \subseteq \Gamma)$ **do** $\Omega := \Omega - \{X \rightarrow Y\};$ $\Gamma := \Gamma \cup Y;$ **return** Γ ;

Case | FD | Entailment | **Equivalence** 1 2 3 4

Example

Algorithm 1: Attribute Closure Algorithm**input:** S, Σ **output:** S^+ **begin** $\Omega := \Sigma$; *// unused* $\Gamma := S$; *// closure***while** $(X \rightarrow Y)$ and $(X \subseteq \Gamma)$ **do** $\Omega := \Omega - \{X \rightarrow Y\}$; $\Gamma := \Gamma \cup Y$;**return** Γ ;**Question**

- $R = \{A, B, C, D\}$
- $\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$

Compute $\{C\}^+$.

+ -

Case | FD | Entailment | **Equivalence** 1 2 3 4

Definition

Equivalence of Set of Functional Dependencies

Two sets of functional dependencies Σ_1 and Σ_2 are **equivalent** if and only if they have the same closure.

$$\Sigma_1 \equiv \Sigma_2 \iff \Sigma_1^+ = \Sigma_2^+$$

We say that Σ_1 is a **cover** of Σ_2 (also Σ_2 is a cover of Σ_1).

Case | FD | Entailment | **Equivalence** 1 2 3 4

Definition

Equivalence of Set of Functional Dependencies

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Question

How do we check equivalence?

Case | FD | Entailment | **Equivalence** 1 2 3 4

Definition

Equivalence of Set of Functional Dependencies

Two sets of functional dependencies Σ_1 and Σ_2 are **equivalent** if and only if they have the same closure.

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We say that Σ_1 is a **cover** of Σ_2 (also Σ_2 is a cover of Σ_1).

Question

How do we check equivalence?

Answer

- For each $(X \rightarrow Y) \in \Sigma_1$, $\Sigma_2 \models (X \rightarrow Y)$ **and**
For each $(X \rightarrow Y) \in \Sigma_2$, $\Sigma_1 \models (X \rightarrow Y)$
(i.e., $Y \subseteq \text{Algo1}(X, \Sigma)$)
- For each $S \subseteq R$, $\text{Algo1}(S, \Sigma_1) = \text{Algo1}(S, \Sigma_2)$

Basically,

$\text{Algo1}(S, \Sigma)$ is the most important algorithm for FD.

Break



Minimal Cover ❶ ❷ ❸ ❹ ❺ ❻ | Exercise | The Chase

Definition

Minimal

A set of functional dependency Σ is **minimal** if and only if

1. The right-hand side of every dependency in Σ is **minimal**.

Every functional dependency is of the form $X \rightarrow \{A\}$.

2. The left-hand side of every dependency in Σ is **minimal**.

For every functional dependency of the form $X \rightarrow \{A\}$,

there is **no** functional dependency $Y \rightarrow \{A\}$ in Σ^+ such that $Y \subset X$.

3. The set itself is **minimal**.

None of the functional dependencies in Σ can be derived from other functional dependencies in Σ

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Examples

Minimal

A set of functional dependency Σ is **minimal** if and only if

1. The **right-hand side** of every dependency in Σ is **minimal**.

Every functional dependency is of the form $X \rightarrow \{A\}$.

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3. The **set** itself is **minimal**.

None of the functional dependencies in Σ can be derived from other functional dependencies in Σ

Question

Consider $R = \{A, B, C\}$ with $\Sigma = \{ \{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\}, \{A\} \rightarrow \{C\} \}$. Which one is a **minimal cover** of Σ ?

Choice	Comment	
A $\{ \{A\} \rightarrow \{B\}, \{A,B\} \rightarrow \{C\} \}$	NO: violated condition 2	✗
B $\{ \{A\} \rightarrow \{B,C\}, \{A\} \rightarrow \{C\} \}$	NO: violated condition 1	✗
C $\{ \{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\} \}$	YES: minimal and cover	✓
D $\{ \{B\} \rightarrow \{C\}, \{A\} \rightarrow \{C\} \}$	NO: not a cover	✗

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Algorithm

Algorithm 2: Minimal Cover Algorithm

input: Σ

output: Minimal Cover of Σ

1. Simplify (*minimize*) the right-hand side of every σ in Σ .
Decompose each $(X \rightarrow Y) \in \Sigma$ into $(X \rightarrow \{A\})$ for each $A \in Y$.
2. Simplify (*minimize*) the left-hand side of every σ in Σ .
For each $(X \rightarrow \{A\})$, if $\Sigma \models (X - \{B\} \rightarrow \{A\})$ for some $B \in X$,
then **replace** $(X \rightarrow \{A\})$ with $((X - \{B\}) \rightarrow \{A\})$.
3. Simplify (*minimize*) the set Σ .
For each $(X \rightarrow \{A\})$, if $(\Sigma - (X \rightarrow \{A\})) \models (X \rightarrow \{A\})$, then **remove** $(X \rightarrow \{A\})$ from Σ .

Important!

The simplification has to be done in **the given order**.

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Canonical

Compact

A set of functional dependencies Σ is **compact**

if and only if

there is no different functional dependencies with the same left-hand side.

Example

$\Sigma = \{ \{A\} \rightarrow \{B\}, \{A\} \rightarrow \{C\} \}$ is **not** compact.

$\Sigma = \{ \{A\} \rightarrow \{B,C\} \}$ is compact.

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Canonical

Compact

A set of functional dependencies Σ is **compact**

if and only if

there is no different functional dependencies with the same left-hand side.

Example

$\Sigma = \{ \{A\} \rightarrow \{B\}, \{A\} \rightarrow \{C\} \}$ is **not** compact.

$\Sigma = \{ \{A\} \rightarrow \{B,C\} \}$ is compact.

Compact and Minimal

A set of functional dependencies Σ is **compact** and **minimal** (*i.e., canonical*)

if and only if

1. The set is compact.
2. The left-hand side of every dependency in Σ is **minimal**.
3. The set itself is **minimal**.

This is also known as **canonical cover**.

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Theorems

Theorem

Every set of functional dependency Σ has at least one minimal cover.

Theorem

Every set of functional dependency Σ has at least one compact cover.

Theorem

Every set of functional dependency Σ has at least one canonical cover.

Minimal Cover 1 2 3 4 5 6 | Exercise | The Chase

Theorems

Theorem

Every set of functional dependency Σ has at least one minimal cover.

Theorem

Every set of functional dependency Σ has at least one compact cover.

Theorem

Every set of functional dependency Σ has at least one canonical cover.

Proof?

The proof is **trivial** as follows.

- | | |
|------------------|---|
| Minimal | Run Algorithm #2 on Σ . The output could be Σ itself. |
| Compact | Apply union rule to the applicable functional dependency in Σ . |
| Canonical | Run Algorithm #2 and apply union rule to the applicable functional dependency. |

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A,B\} \rightarrow \{C,D,E\}, \{A,C\} \rightarrow \{B,D,E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

Singleton

$$\{A\}^+ = \{A\}$$

$$\{B\}^+ = \{B,C,D,E\}$$

$$\{C\}^+ = \{B,C,D,E\}$$

$$\{D\}^+ = \{D\}$$

$$\{E\}^+ = \{E\}$$

Candidate Keys?

None found for singleton.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

Pair

$$\{A, B\}^+ = \{A, B, C, D, E\}$$

$$\{A, C\}^+ = \{A, B, C, D, E\}$$

$$\{A, D\}^+ = \{A, D\}$$

$$\{A, E\}^+ = \{A, E\}$$

$$\{B, C\}^+ = \{B, C, D, E\}$$

$$\{B, D\}^+ = \{B, C, D, E\}$$

$$\{B, E\}^+ = \{B, C, D, E\}$$

$$\{C, D\}^+ = \{B, C, D, E\}$$

$$\{C, E\}^+ = \{B, C, D, E\}$$

$$\{D, E\}^+ = \{D, E\}$$

Candidate Keys?

$$\{A, B\}$$

$$\{A, C\}$$

Any superset of candidate keys will be superkey but not candidate key.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

Triplet

$$\{A, D, E\}^+ = \{A, D, E\}$$

$$\{B, C, D\}^+ = \{B, C, D, E\}$$

$$\{B, C, E\}^+ = \{B, C, D, E\}$$

$$\{B, D, E\}^+ = \{B, C, D, E\}$$

$$\{C, D, E\}^+ = \{B, C, D, E\}$$

Candidate Keys?

$\{A, B\}$
 $\{A, C\}$

*None found for triplet.
Note that we omit all superset of
 $\{A, B\}$ and $\{A, C\}$.*

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A,B\} \rightarrow \{C,D,E\}, \{A,C\} \rightarrow \{B,D,E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

Quadruplet

$$\{B,C,D,E\}^+ = \{B,C,D,E\}$$

Candidate Keys?

$\{A,B\}$
 $\{A,C\}$

*None found for quadruplet.
No need to check quintuplet.*

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A,B\} \rightarrow \{C,D,E\}, \{A,C\} \rightarrow \{B,D,E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

$\Sigma = \{$

$$\{A,B\} \rightarrow \{C,D,E\}$$

$$\{A,C\} \rightarrow \{B,D,E\}$$

$$\{B\} \rightarrow \{C\}$$

$$\{C\} \rightarrow \{B\}$$

$$\{C\} \rightarrow \{D\}$$

$$\{B\} \rightarrow \{E\}$$

$$\{C\} \rightarrow \{E\}$$

$\}$

Step

Start with Σ ,
simplify **right-hand side**.
This step is easy.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A,B\} \rightarrow \{C,D,E\}, \{A,C\} \rightarrow \{B,D,E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

$$\{B\} \rightarrow \{C\}$$

$$\{C\} \rightarrow \{B\}$$

$$\{C\} \rightarrow \{D\}$$

$$\{B\} \rightarrow \{E\}$$

$$\{C\} \rightarrow \{E\}$$

$$\}$$

$$\{A,B\} \rightarrow \{C\}$$

$$\{A,B\} \rightarrow \{D\}$$

$$\{A,B\} \rightarrow \{E\}$$

$$\{A,C\} \rightarrow \{B\}$$

$$\{A,C\} \rightarrow \{D\}$$

$$\{A,C\} \rightarrow \{E\}$$

Step

From simplified RHS,
simplify **left-hand side**.
This step is hard.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A,B\} \rightarrow \{C,D,E\}, \{A,C\} \rightarrow \{B,D,E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

$$\{B\} \rightarrow \{C\}$$

$$\{C\} \rightarrow \{B\}$$

$$\{C\} \rightarrow \{D\}$$

$$\{B\} \rightarrow \{E\}$$

$$\{C\} \rightarrow \{E\}$$

$$\}$$

$\{A,B\} \rightarrow \{C\}$	$\{B\} \rightarrow \{C\}$
$\{A,B\} \rightarrow \{D\}$	$\{B\} \rightarrow \{D\}$
$\{A,B\} \rightarrow \{E\}$	$\{B\} \rightarrow \{E\}$
$\{A,C\} \rightarrow \{B\}$	$\{C\} \rightarrow \{B\}$
$\{A,C\} \rightarrow \{D\}$	$\{C\} \rightarrow \{D\}$
$\{A,C\} \rightarrow \{E\}$	$\{C\} \rightarrow \{E\}$

Step

From simplified LHS,
simplify the **set**.
This step is medium.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

$$\{B\} \rightarrow \{C\}$$

$$\{C\} \rightarrow \{B\}$$

$$\{C\} \rightarrow \{D\}$$

$$\{B\} \rightarrow \{E\}$$

$$\{C\} \rightarrow \{E\}$$

$$\}$$

~~$$\{B\} \rightarrow \{C\}$$~~ (duplicate)

$$\{B\} \rightarrow \{D\}$$

~~$$\{B\} \rightarrow \{E\}$$~~ (duplicate)

~~$$\{C\} \rightarrow \{B\}$$~~ (duplicate)

~~$$\{C\} \rightarrow \{D\}$$~~ (duplicate)

~~$$\{C\} \rightarrow \{E\}$$~~ (duplicate)

Step

Remove duplicates.

This part is easy.

Next is to find transitive.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

~~$\{B\} \rightarrow \{D\}$~~

from $\{B\} \rightarrow \{C\}$ and $\{C\} \rightarrow \{D\}$

$\{B\} \rightarrow \{C\}$

$\{C\} \rightarrow \{B\}$

$\{C\} \rightarrow \{D\}$

~~$\{B\} \rightarrow \{E\}$~~

from $\{B\} \rightarrow \{C\}$ and $\{C\} \rightarrow \{E\}$

$\{C\} \rightarrow \{E\}$

 $\}$

Step

Remove transitives.

The remainder is one possible **minimal cover**.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

$$\begin{aligned} &\{B\} \rightarrow \{C\} \\ &\{C\} \rightarrow \{B\} \\ &\{C\} \rightarrow \{D\} \\ &\{C\} \rightarrow \{E\} \end{aligned}$$

}

 $\Sigma = \{$

$$\begin{aligned} &\{B\} \rightarrow \{C\} \\ &\{B\} \rightarrow \{D\} \\ &\{B\} \rightarrow \{E\} \\ &\{C\} \rightarrow \{B\} \end{aligned}$$

}

 $\Sigma = \{$

$$\begin{aligned} &\{B\} \rightarrow \{C\} \\ &\{B\} \rightarrow \{D\} \\ &\{C\} \rightarrow \{B\} \\ &\{C\} \rightarrow \{E\} \end{aligned}$$

}

Note

Other minimal covers are possible.

They can be achieved by considering the constraints in different order.

Some minimal cover may be **unreachable** using this algorithm.

Minimal Cover | **Exercise** 1 2 3 4 5 6 7 8 9 10 | The Chase

Exercise

Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

1. Find the **candidate keys** of R with Σ
2. Compute the **minimal cover** of Σ
3. Compute the **canonical cover** of Σ

 $\Sigma = \{$

$$\begin{array}{l} \{B\} \rightarrow \{C\} \\ \{C\} \rightarrow \{B, D, E\} \end{array}$$

 $\}$ $\Sigma = \{$

$$\begin{array}{l} \{B\} \rightarrow \{C, D, E\} \\ \{C\} \rightarrow \{B\} \end{array}$$

 $\}$ $\Sigma = \{$

$$\begin{array}{l} \{B\} \rightarrow \{C, D\} \\ \{C\} \rightarrow \{B, E\} \end{array}$$

 $\}$

Note

Apply **union** rule.

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Preliminary

The Chase

Let Σ be a set of functional dependencies* on a relation schema R . **The Chase** is an algorithm that solves the decision problem of whether the functional dependency σ is satisfied by R with Σ .

in other words

$$(R \text{ with } \Sigma) \models \sigma$$

*Also work for multi-valued dependencies, but we will omit them unless time permits.

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C,D\} \rightarrow \{A,B\}$?

Setup

Create an instance r of relation schema R with **two tuples** and **distinct values** for all attributes.

A	B	C	D
α	α	α	α
a_1	b_1	c_1	d_1

*The first row is a row of **distinguished variables**.

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Initialization

We want to chase $\{C, D\} \rightarrow \{A, B\}$. Make all $\{C, D\}$ -values the same.

A	B	C	D
α	α	α	α
a_1	b_1	c_1	d_1

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C,D\} \rightarrow \{A,B\}$?

Initialization

We want to chase $\{C,D\} \rightarrow \{A,B\}$. Make all $\{C,D\}$ -values the same.

A	B	C	D
α	α	α	α
a_1	b_1	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Chasing

Use $\{C\} \rightarrow \{A\}$. For each pair of tuple with the same $\{C\}$ -values, make their $\{A\}$ -values the same.

A	B	C	D
α	α	α	α
a_1	b_1	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Chasing

Use $\{C\} \rightarrow \{A\}$. For each pair of tuple with the same $\{C\}$ -values, make their $\{A\}$ -values the same.

A	B	C	D
α	α	α	α
α	b_1	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Chasing

Use $\{A\} \rightarrow \{B\}$. For each pair of tuple with the same $\{A\}$ -values, make their $\{B\}$ -values the same.

A	B	C	D
α	α	α	α
α	b_1	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C, D\} \rightarrow \{A, B\}$?

Chasing

Use $\{A\} \rightarrow \{B\}$. For each pair of tuple with the same $\{A\}$ -values, make their $\{B\}$ -values the same.

A	B	C	D
α	α	α	α
α	α	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C,D\} \rightarrow \{A,B\}$?

The End

There is nothing else to do. We observe that the table satisfies $\{C,D\} \rightarrow \{A,B\}$.

A	B	C	D
α	α	α	α
α	α	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Functional Dependency

Question

Let

$$R = \{A, B, C, D\}$$

$$\Sigma = \{ \{A\} \rightarrow \{B\}, \{C\} \rightarrow \{A\} \}$$

Is $\Sigma \models \{C,D\} \rightarrow \{A,B\}$?

Important

r also satisfies $\{B\} \rightarrow \{A\}$. But this is a **coincidence**. We can only answer question about $\{C,D\} \rightarrow \{A,B\}$.

A	B	C	D
α	α	α	α
α	α	α	α

Minimal Cover | Exercise | **The Chase** 1 2 3 4 5 6 7 8

Theorems

Theorem

The Chase is **sound** and **complete** for σ .

$r \models \sigma$ is equivalent to $\Sigma \models \sigma$

Theorem

The Chase always terminates for functional dependencies*.

Theorem

The Chase always produce **counter-example** if it exists for functional dependencies*.

Lossless-Join

We will use The Chase to check lossless join decomposition.

*Also for multi-valued dependencies.

Appendix

Completeness

Proof

Completeness of Armstrong's Axioms (Contrapositive)

$$(\Sigma \not\models X \rightarrow Y) \Rightarrow (\Sigma \not\models X \rightarrow Y)$$

Proof

1. We know that $\Sigma \not\models X \rightarrow \{A\}$ (can be generalized to Y).
2. Therefore, $\{A\} \notin X$ (otherwise, can be proven using Reflexivity).
3. Construct instance r of R such that $S = \{B \mid \Sigma \vdash X \rightarrow \{B\}\}$.

S			R - S		
1	...	1	1	...	1
1	...	1	0	...	0

By construction $\{A\} \notin S$.

It has two tuples, they differ in $R - S$ attributes.

4. Assume there exists $\Sigma \vdash V \rightarrow W$ such that $r \not\models V \rightarrow W$.
 - A. Therefore, $V \subseteq S$. Otherwise, $r \models V \rightarrow W$ (i.e., we can violate the dependency).
 - B. Therefore, $\Sigma \vdash X \rightarrow V$ by construction of S and by Union rule.
 - C. Therefore, $\Sigma \vdash X \rightarrow W$ by transitivity of (B) and (4).
 - D. Therefore, $W \subseteq S$ by construction of S .
 - E. Therefore, $r \models V \rightarrow W$ since $(V \cup W) \subseteq S$ (due to (A) and (D)).
 - F. This is a **contradiction**. Hence, if $\Sigma \vdash V \rightarrow W$, then $r \models V \rightarrow W$.

Completeness

Proof

Completeness of Armstrong's Axioms (Contrapositive)

$$(\Sigma \not\models X \rightarrow Y) \Rightarrow (\Sigma \not\models X \rightarrow Y)$$

Proof

5. Therefore, if $\Sigma \vdash X \rightarrow Y$, then $r \models X \rightarrow Y$.
6. We know that $\forall (X \rightarrow Y) \in \Sigma : (\Sigma \vdash X \rightarrow Y)$ (*the empty proof*).
7. Therefore $r \models X \rightarrow Y$ by (5).
8. We observe that $r \models X \rightarrow \{A\}$ by construction (*verify that $A \notin S$*).
9. Therefore, $X \rightarrow \{A\} \notin \Sigma^+$ by definition of entailment due to (7) and (8).

Hence, all instances that verify the functional dependencies in Σ must verify $X \rightarrow \{A\}$ for it to be entailed by Σ . We constructed one that does not.

```
postgres=# exit
```

```
Press any key to continue . . .
```